

Nonpungent capsaicin analogs (capsinoids) increase energy expenditure through the activation of brown adipose tissue in humans.

BACKGROUND: Capsinoids-nonpungent capsaicin analogs-are known to activate brown adipose tissue (BAT) thermogenesis and whole-body energy expenditure (EE) in small rodents. BAT activity can be assessed by [^{18}F]fluorodeoxyglucose-positron emission tomography (FDG-PET) in humans. **OBJECTIVES:** The aims of the current study were to examine the acute effects of capsinoid ingestion on EE and to analyze its relation to BAT activity in humans. **DESIGN:** Eighteen healthy men aged 20-32 y underwent FDG-PET after 2 h of cold exposure (19°C) while wearing light clothing. Whole-body EE and skin temperature, after oral ingestion of capsinoids (9 mg), were measured for 2 h under warm conditions (27°C) in a single-blind, randomized, placebo-controlled, crossover design. **RESULTS:** When exposed to cold, 10 subjects showed marked FDG uptake into adipose tissue of the supraclavicular and paraspinal regions (BAT-positive group), whereas the remaining 8 subjects (BAT-negative group) showed no detectable uptake. Under warm conditions (27°C), the mean ($\pm$ SEM) resting EE was 6114 ± 226 kJ/d in the BAT-positive group and 6307 ± 156 kJ/d in the BAT-negative group (NS). EE increased by 15.2 ± 2.6 kJ/h in 1 h in the BAT-positive group and by 1.7 ± 3.8 kJ/h in the BAT-negative group after oral ingestion of capsinoids ($P < 0.01$). Placebo ingestion produced no significant change in either group. Neither capsinoids nor placebo changed the skin temperature in various regions, including regions close to BAT deposits. **CONCLUSION:** Capsinoid ingestion increases EE through the activation of BAT in humans. This trial was registered at <http://www.umin.ac.jp/ctr/> as UMIN 000006073.